are unable to get the answer to this problem, please download and execute the script *pstricks6.tex*.

Loops in PSTricks

You are tasked with developing the network topology diagram of your office LAN. The diagram uses a circle to denote a computer in your office LAN and you are working in a large company with hundreds of computers. How will you draw the circles in your diagram? Well, you don't have to draw each and every one of those circles manually. PSTricks offers very powerful looping and control structures to help you draw a large number of similar images very easily. Consider the LaTeX file *pstricks7.tex* shown below:

```
\documentclass{article}
\usepackage{multido}
\usepackage{pstricks}
\begin{document}
  \begin{pspicture}(10,10)
    \multido{\ia=0+1}{3}
    {
     \multido{\ib=0+1}{3}
     {
         \rput(\ia,\ib){\pscircle[linecolor=red,fillstyle=sol
id,fillcolor=blue]{0.4}}
     }
    }
  \end{pspicture}
\end{document}
```

Let us go through the code in detail to understand PSTricks loops better. There are different ways to use loops in PSTricks, but we will discuss only one such technique which uses a macro called *multido*. The line of code '\usepackage{multido}' allows us to use the functionalities of the macro *multido*. The line of code '\multido{\ia=0+1}{3}' defines an outer loop that gets executed three times. Similarly, the line of code '\multido{\ib=0+1}{3}' defines an inner loop that gets executed three times. So the line of code '\rput(\ia,\ib){\pscircle[linecolor=red,fillstyle=solid, fillcolor=blue]{0.4}}' inside the inner loop gets executed for a total of nine times. The macro *rput* is used here to set the reference point. In this case, the reference points are specified by the loop variables *ia* and *ib*, both of which are initialised to zero and get incremented by 1. Due to this reason, the centres of the nine circles with radius of 0.4 units drawn by the code fragment 'pscircle[linecolor =red,fillstyle=solid,fillcolor=blue]{0.4}' are different. On executing the script, *pstricks7.tex* will produce the image shown in Figure 6 as output.

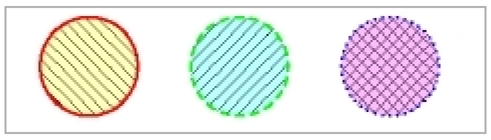

Figure 5: Loops in PSTricks

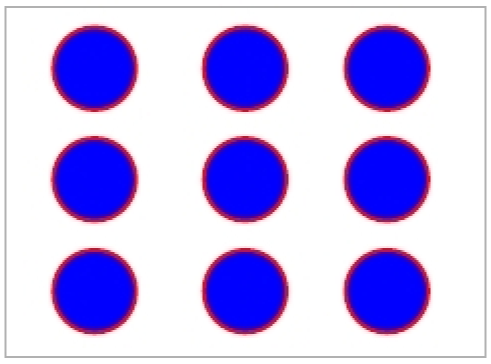

Figure 6: Different styles in PSTricks

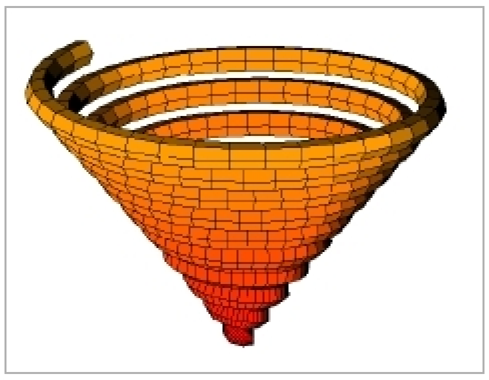

Figure 7: A 3D plot with PSTricks

Now let us make a small modification to the file *pstricks7.tex* to have a glimpse of the real power of PSTricks. The line of code:

```
\multido{\ia=0+1}{3}
```

...is replaced with the line of code:

```
\multido{\ia=0+1}{10}
```

...and the line of code:

```
\multido{\ib=0+1}{3}
```

...is replaced with the line of code:

```
\multido{\ib=0+1}{10}
```

... in *pstricks7.tex* to obtain the file *pstricks8.tex*. On execution, the script *pstricks8.tex* will produce an image containing 100 circles, each with a radius of 0.4 units. All we did to change an image showing nine circles to an image showing 100 circles is modify just two parameters in maybe 30 seconds! Imagine how much time would have been taken by normal image manipulation software to make a similar change.

So, once you overcome the initial difficulties of learning tools like PSTricks and PGF/TikZ, they will save you a lot of time and effort when compared to conventional image manipulation tools. This is yet another advantage of PSTricks and PGF/TikZ.

One of the most powerful aspects of PSTricks is its ability to produce 3D graphics. Due to the complexity in understanding the topic, it has not been discussed in this article. But to give you an idea about the 3D plotting capabilities of PSTricks, I have included a 3D plot generated by PSTricks. To work with 3D plots in PSTricks, you need to have some additional files installed in your system or placed in the same directory containing your *.tex* file that generates 3D plots. The files are *pst-solides3d.pro*, *pst-solides3d.sty* and *pst-solides3d.tex*. The plot in Figure 7 is obtained from a slightly modified version of a file called *exa062.tex* provided by the official PSTricks website of the TeX Users Group.

If you are really interested in PSTricks, I suggest that you explore the website *https://tug.org/PSTricks/main.cgi/*, which is full of PSTricks example files and tutorials that will guide you through to mastery. END

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